$$\mathbf{U}^\dagger(\epsilon) = I + i\epsilon\mathbf{H}^\dagger. \qquad (4.7)$$

Now we plug Eqs. 4.6 and 4.7 into the unitarity condition of Eq. 4.5:

$$(I + i\epsilon\mathbf{H}^\dagger)(I - i\epsilon\mathbf{H}) = I.$$

Expanding to first order in ε, we find

$$\mathbf{H}^\dagger - \mathbf{H} = 0$$

or, in a format that is more illuminating,

$$\mathbf{H}^\dagger = \mathbf{H}. \qquad (4.8)$$

This last equation expresses the unitarity condition. But it also says that **H** is a Hermitian operator. This has great significance. We can now say that **H** is an observable, and has a complete set of orthonormal eigenvectors and eigenvalues. As we proceed, **H** will become a very familiar object, namely the *quantum Hamiltonian*. Its eigenvalues are the values that would result from measuring the energy of a quantum system. Exactly why we identify **H** with the classical concept of a Hamiltonian, and its eigenvalues with energy, will become clear shortly.

Let's return now to Eq. 4.1 and specialize it to the infinitesimal case $t = \varepsilon$. Using Eq. 4.6, we find

$$|\Psi(\epsilon)\rangle = |\Psi(0)\rangle - i\epsilon\mathbf{H}|\Psi(0)\rangle.$$

This is just the kind of equation that we can easily turn into a differential equation. First, we transpose the first term on the right side over to the left side, and then divide by ε: